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WEB-BASED ROAD DAMAGE DETECTION AND MAPPING SYSTEM USING YOLOv8 AND GPS SYNCHRONIZATION

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CHAPTER I

INTRODUCTION

Road infrastructure plays a vital role in transportation efficiency, economic activity, and public safety. However, road damage such as potholes, longitudinal cracks, transverse cracks, and alligator cracks continues to negatively affect road quality and transportation reliability. Traditional road inspection methods commonly rely on manual visual surveys and documentation, which are often time-consuming, labor-intensive, and prone to inconsistencies in assessment and reporting. As road networks continue to expand, there is an increasing need for more efficient and organized approaches to road monitoring and maintenance.

This study aims to develop a web-based road damage detection and mapping system to improve the process of identifying and documenting road damage from road survey videos. The proposed system seeks to support more efficient road monitoring workflows, organized maintenance records, and improved accessibility of road condition information through automated detection and mapping functionalities. The study is intended to benefit infrastructure personnel, road maintenance teams, planners, and researchers while contributing to the development of technology-driven solutions for infrastructure management.

1.1 Project Context

Manual road inspection remains one of the commonly used methods for monitoring road conditions and identifying damaged road segments. In this process, personnel physically inspect roads and manually document observed damages such as potholes and cracks. While this approach allows direct human assessment, it is often inefficient for large-scale road monitoring due to the amount of time, labor, and documentation effort required. Manual inspections may

also result in inconsistencies in assessment and delays in maintenance response, particularly when road damage reports lack precise geographic and temporal information.

To address these limitations, researchers have explored the application of computer vision and deep learning technologies for automated road damage detection. One of the most widely used object detection architectures is YOLO (You Only Look Once), introduced by Redmon et al. (2016), which is known for its capability to perform object detection with high speed and accuracy. The availability of large annotated datasets such as RDD2022, containing over 47,000 road damage images from multiple countries (Arya et al., 2022), has further supported the development of highly accurate road damage detection systems. Studies by Pham et al. (2022), Roy and Bhaduri (2023), and Pham et al. (2024) demonstrated the effectiveness of YOLO-based models in detecting road anomalies from images and continuous video feeds.

Several related studies have attempted to apply these technologies within the Philippine context. Sarmiento (2021) explored the use of YOLOv4 for detecting pavement damage using Philippine road images; however, the implementation remained experimental and lacked a deployable monitoring platform. Fortin et al. (2024) developed the EyeWay system capable of pothole mapping, although the system required dedicated hardware devices that may increase deployment costs. Meanwhile, Bustria et al. (2025) developed a YOLOv11-based mobile application for road damage classification, but the system primarily relied on static image uploads rather than continuous road survey video analysis integrated with geospatial mapping.

Unlike previous implementations that focus mainly on image-based detection or hardware-dependent systems, the proposed study integrates YOLOv8-based road damage detection with GPS synchronization and interactive geospatial visualization within a centralized

web application. The study specifically addresses the four road damage categories present in the RDD2022 dataset, namely potholes, longitudinal cracks, transverse cracks, and alligator cracks. Through uploaded road survey videos and GPS track files, the proposed system aims to automate the detection, localization, and mapping of road damage records to support more organized monitoring and maintenance workflows.

The conceptual framework of the study follows an input-process-output structure. The inputs consist of uploaded road survey videos and GPS track files. The process includes frame extraction, YOLOv8-based road damage detection, timestamp extraction, GPS synchronization, and geospatial mapping. The outputs of the system include detected road damage records, geographic coordinates, timestamps, and interactive map visualizations. A system architecture or conceptual framework diagram illustrating this workflow will be presented in the succeeding sections of the study.

1.2 Purpose and Description

This study aims to develop a web-based road damage detection and mapping system utilizing YOLOv8 and GPS synchronization to automate the detection, localization, and mapping of road damage from uploaded road survey videos. The proposed system focuses on identifying multiple road damage types present in the RDD2022 dataset, including potholes, longitudinal cracks, transverse cracks, and alligator cracks. Instead of requiring real-time mobile-device processing, users may upload continuous road survey footage together with GPS track files through a centralized web platform for automated analysis.

The system utilizes a trained YOLOv8 object detection model to process uploaded videos frame-by-frame in order to identify road damage within road surfaces. Once detected, the

application generates bounding boxes around the damaged areas, extracts the corresponding timestamps of the detections, and synchronizes the information with GPS track data to determine the geographic coordinates of the detected road damage. The synchronized data is then presented through an interactive mapping interface, allowing users to visualize road damage locations and corresponding road survey records more efficiently.

By integrating computer vision, video analysis, GPS synchronization, and geospatial mapping technologies within a single web application, the proposed system aims to improve the efficiency, organization, and accessibility of road damage monitoring workflows. The study also seeks to provide a practical and scalable alternative to traditional manual road inspection methods by reducing the need for continuous human visual review and manual documentation processes.

1.3 Statement of the Problem

Current road inspection and monitoring processes remain heavily dependent on manual surveys and visual assessment, making road condition monitoring labor-intensive, time-consuming, and difficult to manage across large road networks (Dabous et al., 2025). Delays and inconsistencies in documenting potholes and pavement cracks may lead to inefficient maintenance planning, increased transportation risks, and prolonged road deterioration. Although recent advancements in artificial intelligence and computer vision introduced automated road damage detection systems using YOLO-based deep learning models, many existing implementations remain limited to standalone object detection and lack integrated GPS synchronization, geospatial visualization, and centralized web-based monitoring functionalities (Pham et al., 2022; Roy & Bhaduri, 2023). Several systems also rely on static image uploads or dedicated mobile hardware, reducing scalability and accessibility for continuous road survey

operations (Fortin et al., 2024; Bustria et al., 2025). Within the Philippine context, additional challenges such as irregular pavement textures, inconsistent lighting conditions, flooding, and varying weather conditions may negatively affect detection reliability, especially for models trained primarily on foreign datasets (Sarmiento, 2021). Furthermore, synchronization inconsistencies between detection outputs and geographic data may reduce the effectiveness of mapping-based road monitoring systems for infrastructure maintenance workflows. These limitations highlight the need for a centralized web-based road damage detection and mapping system capable of integrating YOLOv8-based detection, GPS synchronization, timestamp extraction, and geospatial visualization for more efficient road infrastructure monitoring.

This study addresses the following research questions:

1. Can object detection models (e.g., YOLOv8) effectively detect and localize potholes, and different types of cracks from uploaded road survey videos?

2. In what way can a web application integrate road damage detection data with GPS location data to generate a geospatially-aware damage log?

3. Can GPS tracking synchronization with video timestamps provide accurate mapping of road damage locations for actionable maintenance workflows?

4. How does incorporating both automated video analysis and GPS geolocation improve road inspection efficiency and decision-making for LGUs compared to traditional manual inspection methods?

5. What limitations and challenges arise in developing a geospatially-integrated video-based road damage detection web application, particularly in terms of GPS accuracy, time synchronization, video quality variations, and detection accuracy?

1.4 Hypothesis

The implementation of a web-based road damage detection and mapping system utilizing the YOLOv8 object detection model integrated with GPS synchronization will effectively automate the identification and geographic localization of multiple road damage types from uploaded road survey videos. The proposed system is expected to improve the efficiency and accuracy of road inspection processes by reducing reliance on manual visual inspections, generating geospatially-aware damage records, and supporting faster maintenance planning and decision-making compared to traditional road assessment methods.

1.5 Research Objectives

1.5.1 General Objective

To develop a web-based road damage detection and mapping system utilizing YOLOv8 object detection and GPS synchronization that automatically analyzes uploaded road survey videos to detect road damage, generate timestamps, and map geographic coordinates for organized road monitoring workflows.

1.5.2 Specific Objectives

1. To design and develop a web-based user interface that allows users to upload road survey videos and corresponding GPS track files in GPX or CSV format.

2. To train and integrate a YOLOv8-based object detection model optimized for frame-by-frame road damage detection using datasets such as RDD2022 and locally collected road images.
3. To implement a frame-analysis feature that generates bounding boxes around detected road damage, extracts detection timestamps, and synchronizes the information with GPS track data to determine geographic coordinates.
4. To implement interactive map visualization features using open-source geospatial tools such as React-Leaflet and OpenStreetMap for displaying detected road damage locations and GPS track data.
5. To evaluate the system's performance in terms of detection accuracy, GPS synchronization accuracy, processing efficiency, and overall usability for road monitoring workflows.

1.6 Significance of the Study

The study focuses on the development of a web-based road damage detection and mapping system that integrates YOLOv8 object detection, GPS synchronization, and geospatial visualization to automate the monitoring of road conditions from uploaded road survey videos. By processing detected road damages together with timestamp and geographic coordinate data, the system aims to provide a more organized and accessible approach to infrastructure monitoring. The study benefits the following.

Road Maintenance Personnel. The system assists road maintenance personnel by automating the identification and mapping of potholes and road cracks from uploaded survey videos. Through synchronized GPS data and generated road damage records, maintenance teams

can more efficiently locate damaged road segments, prioritize repair operations, and reduce the time required for manual inspection and documentation.

Local Government Units (LGUs). Local government units may utilize the system to support infrastructure monitoring and maintenance planning. The centralized and mapped road damage records can improve decision-making processes related to repair scheduling, budgeting, and resource allocation, while also improving accessibility to organized road condition data.

Infrastructure Planners and Engineers. Infrastructure planners and engineers may benefit from the geospatial visualization and synchronized road condition information produced by the system. The generated geographic data can support planning activities, infrastructure assessments, and the monitoring of road deterioration patterns for more informed infrastructure management.

Motorists and the General Public. The study indirectly benefits motorists and commuters by supporting faster identification and monitoring of damaged road segments. Improved maintenance response and organized monitoring workflows may contribute to safer and more reliable transportation conditions by reducing risks associated with deteriorated roads.

Researchers and Developers. The proposed system may serve as a reference for researchers and developers interested in computer vision, geospatial systems, and intelligent transportation technologies. The study contributes to the growing application of AI-based monitoring systems for infrastructure management and may support future developments involving automated road inspection and smart city technologies.

Academic Institutions and Students. Academic institutions and students may use the study as a learning reference in areas related to artificial intelligence, computer vision, geospatial

technologies, and web-based systems. The study demonstrates the practical application of machine learning and mapping technologies in solving real-world infrastructure problems.

Future Researchers. The study provides a foundation for future research involving automated infrastructure monitoring systems. Future researchers may expand the study by integrating real-time roadside detection, drone-based inspection systems, additional road damage classifications, and localized datasets for improving compatibility with Philippine road environments.

1.7 Scope and Limitations

This study focuses on the development of a web-based road damage detection and mapping system utilizing YOLOv8 and GPS synchronization. The system is designed to detect and localize multiple road damage types from uploaded road survey videos through frame-by-frame video analysis and geospatial mapping functionalities. The study specifically covers the four road damage categories present in the RDD2022 dataset, namely potholes, longitudinal cracks, transverse cracks, and alligator cracks. Uploaded videos are processed using the YOLOv8 object detection model to identify road damage, generate bounding boxes, extract timestamps, and synchronize detections with GPS track data for interactive map visualization using open-source geospatial technologies such as React-Leaflet and OpenStreetMap.

The proposed application is designed as a centralized web-based platform and performs processing asynchronously after video uploads are completed; therefore, the system does not support real-time roadside detection or live mobile-device inference. The detection model is trained primarily using the RDD2022 dataset supplemented with locally collected road images to improve compatibility with Philippine road conditions.

The system is not intended to replace current road inspection workflows because its performance may still be limited by several factors, including poor lighting conditions, unstable camera movement, low-resolution video footage, weather conditions, GPS inaccuracies, and synchronization inconsistencies between video timestamps and GPS track data. In addition, variations in road appearance, camera angles, and environmental conditions may affect the consistency of detecting different types of road damage. Despite these limitations, the system may still improve the efficiency and organization of road inspection processes by supporting data collection and monitoring activities alongside manual validation and engineering assessment during maintenance planning and decision-making processes.

CHAPTER II

REVIEW OF RELATED LITERATURE

Road infrastructure plays an important role in transportation efficiency, economic activity, and public safety. However, maintaining roads remains a challenge for many cities and municipalities because traditional inspection methods rely heavily on manual surveys, visual observation, and written reports. These methods are often time-consuming, costly, and inconsistent, especially in areas with limited manpower and resources. Because of these challenges, researchers and developers have explored the use of artificial intelligence, computer vision, and deep learning technologies to automate the detection of road damage such as potholes, longitudinal cracks, transverse cracks, and alligator cracks.

Recent developments in deep learning, particularly the YOLO (You Only Look Once) architecture introduced by Joseph Redmon et al. (2016), have significantly improved real-time object detection systems. Modern YOLO versions such as YOLOv8 are now widely used in pavement analytics because of their speed, accuracy, and ability to detect irregular road defects from images and continuous road survey videos. Researchers have also integrated these detection systems with GPS synchronization and web-based platforms to improve road monitoring efficiency and accessibility of infrastructure records.

This chapter presents related literature and studies that support the development of the proposed web-based road damage detection and mapping system. The discussion is divided into foreign and local studies to provide both international and Philippine perspectives regarding automated road monitoring systems. The chapter also discusses advancements in YOLO-based object detection, geospatial technologies, transfer learning, and centralized web-based

infrastructures that contribute to the feasibility of the proposed study. Lastly, the synthesis summarizes the common findings, research gaps, and relevance of the reviewed studies to the present research.

2.1 Foreign Literature and Studies

The development of automated road damage detection systems emerged from the increasing need for more efficient infrastructure monitoring solutions. Traditional road inspections relied mainly on manual surveys conducted by maintenance personnel and engineers. Although manual inspection allows direct assessment of road conditions, it often becomes inefficient for large-scale road networks because of high labor requirements, long inspection times, and inconsistent reporting processes. Due to these limitations, researchers began exploring artificial intelligence and computer vision technologies for automated pavement analysis.

One of the most significant developments in computer vision was introduced by Joseph Redmon et al. (2016) through the YOLO framework. Their study proposed a unified real-time object detection architecture capable of detecting objects directly from images with high speed and accuracy. YOLO became highly influential because it simplified object detection pipelines while maintaining competitive detection performance, making it suitable for real-time applications such as transportation monitoring and road inspection systems.

As object detection architectures evolved, researchers improved YOLO models for more complex road environments. Pham, Pham, and Dang (2022) implemented a real-time road damage detection system using deep learning-based image analysis. Their study demonstrated that YOLO-based models could accurately identify road defects such as potholes and cracks from captured road images. Similarly, Roy and Bhaduri (2023) utilized YOLOv8 for real-time

road damage detection and showed that the newer architecture improved detection performance and processing efficiency compared to previous YOLO versions.

Further advancements in YOLO architectures were explored by Pham et al. (2024), who analyzed successive improvements in YOLO models for detecting road anomalies. Their findings showed that newer YOLO versions improved feature extraction, multi-scale detection, and robustness under varying environmental conditions such as shadows, lighting inconsistencies, and road surface variations. These improvements contributed to more accurate pavement analysis systems.

One of the most widely adopted models today is YOLOv8. According to Irsal et al. (2025), YOLOv8 introduced an anchor-free detection mechanism that improved the localization of potholes and irregular road damage. Their study demonstrated that YOLOv8 could process road imagery in real time while maintaining high detection precision. Yin et al. (2024) further validated the effectiveness of YOLOv8 by applying the model to detect potholes, sewer covers, and manholes. Their research achieved high mean Average Precision (mAP) results through enhanced multi-scale feature learning techniques.

Researchers also explored modified YOLOv8 architectures for improving road damage detection reliability. Zeng and Zhong (2024) developed the YOLOv8-PD model, which enhanced feature fusion mechanisms to reduce false detections and improve pavement defect recognition. Likewise, Zhang et al. (2025) proposed the OBC-YOLOv8 model that integrated attention-enhanced feature learning into the YOLOv8 architecture. Their findings revealed improved detection of small and distant potholes commonly observed in dashcam footage and continuous road survey videos.

The growing success of AI-based road inspection systems also emphasized the importance of large and diverse training datasets. Arya et al. (2022) introduced the RDD2022 dataset, a multinational collection of over 47,000 annotated road damage images from several countries. The dataset allowed deep learning models to learn from different pavement textures, environmental conditions, and road damage patterns, significantly improving generalized detection performance.

Similarly, Asamoah et al. (2025) developed the PaveSync dataset, a unified and comprehensive dataset for pavement distress analysis and classification. Their research demonstrated that combining multiple datasets improves benchmarking reliability and strengthens model robustness across varying geographic conditions. These datasets became essential resources for training modern road damage detection systems.

Another major advancement in infrastructure monitoring is the integration of centralized web-based systems for automated road analysis. According to Dabous et al. (2025), web-based AI platforms improve scalability and accessibility by utilizing centralized GPU processing for intensive frame-by-frame video analysis. Centralized systems reduce the hardware limitations commonly encountered in mobile devices during continuous deep learning operations and improve the organization of road monitoring records.

The importance of centralized processing systems was also emphasized by Dabous et al. (2025), whose systematic review explained that web-based AI platforms provide access to high-performance GPUs capable of intensive frame-by-frame processing. The researchers noted that mobile devices often experience overheating, battery drainage, and inconsistent performance

during continuous deep learning tasks, making centralized web processing a more efficient solution.

Modern road inspection systems also integrate geospatial technologies to improve infrastructure management. Busireddy et al. (2026) combined deep learning detection with GPS synchronization methods to improve pothole localization accuracy. Their findings showed that synchronizing detection timestamps with GPS track data reduced the time required for maintenance teams to locate damaged road segments.

Albouchi et al. (2025) also highlighted the usefulness of geospatial visualization through heat map technologies. Their enhanced YOLO-based monitoring system allowed infrastructure planners to prioritize maintenance operations based on the concentration of road damage in specific locations. The study demonstrated how mapping technologies improve infrastructure decision-making and maintenance planning.

Transfer learning also became an important component in improving road damage detection performance across different regions. Alwahhab et al. (2025) demonstrated that fine-tuning pre-trained models using localized datasets significantly reduced false detections caused by shadows, wet pavements, and environmental inconsistencies. Their findings proved that locally adapted deep learning models achieve higher accuracy in developing countries with varying road conditions.

In addition, the Asian Development Bank (2026) emphasized that machine learning technologies are essential for achieving sustainable infrastructure monitoring in developing nations. Their guidebook encouraged the use of AI-based road quality monitoring systems

because of their scalability, cost efficiency, and ability to support data-driven maintenance operations.

2.2 Local Literature and Studies

In the Philippine context, road damage detection remains an emerging field because of the country's unique infrastructure and environmental challenges. Philippine roads are frequently affected by heavy rainfall, flooding, overloaded vehicles, and inconsistent maintenance practices, resulting in rapid pavement deterioration. Because of these conditions, manual road inspection methods often become inefficient and unable to provide timely, organized, and accurate monitoring of road conditions. Recent local studies highlight the growing need for automated and localized road damage detection systems tailored to Philippine environments.

One of the earliest local studies related to automated pavement analysis was conducted by Sarmiento (2021), who applied YOLOv4 and DeepLabv3 for pavement distress detection using locally collected Philippine road images. The study revealed that foreign-trained models struggled to adapt to Philippine road conditions due to irregular asphalt textures, shadows, debris, and inconsistent lighting conditions commonly present on local roads. This finding emphasized the importance of developing localized datasets and adapting AI models specifically for Philippine infrastructure environments.

The importance of localized deep learning systems was further explored by Sarmiento and Reyes (2022), who developed a deep learning-based road damage classification system for Philippine highways. Their research utilized convolutional neural networks to classify potholes and road cracks using locally trained datasets. The researchers concluded that models trained on Philippine road data achieved improved detection performance and classification accuracy

compared to systems relying on foreign datasets, reinforcing the significance of localized model development.

Another significant local study was conducted by Manalo et al. (2022), who developed a transfer learning-based pothole detection system for Philippine roads using YOLOv3 and a locally compiled pothole image dataset. The project, developed through collaboration among Batangas State University, Adamson University, and FEU Institute of Technology, achieved an average precision of 95.43%. However, the accuracy of detecting individual potholes varied due to differences in pothole shapes and sizes commonly found on Philippine roads. The findings demonstrated the feasibility of utilizing pre-trained deep learning models with limited local datasets while highlighting the challenges of inconsistent pavement conditions.

As computer vision technologies advanced, local researchers also explored the integration of geospatial technologies and mapping systems to improve road monitoring workflows. Fortin et al. (2024) developed the EyeWay pothole mapping system, which combined computer vision and GPS technologies for pothole localization and mapping. Their findings showed that geospatial technologies significantly improved the accuracy of pothole location tracking and road monitoring operations. The study supports the integration of GPS synchronization and mapping functionalities in automated road damage detection systems.

At the policy and infrastructure level, Dacanay et al. (2024) conducted an assessment of road condition monitoring practices and technologies in the Philippines under the DOST-funded Project PAVE at UP Diliman. The researchers reviewed existing DPWH road monitoring practices and compared them with international automated systems. Their findings revealed that the Philippines still heavily relies on manual road inspections and that many foreign automated

monitoring systems are not directly applicable to Philippine road environments because of local infrastructure conditions and resource limitations. The study emphasized the need for customized automated road survey systems specifically designed for Philippine road infrastructure.

In another local study, Ramos et al. (2024) developed a localized Pavement Condition Index (PCI) model for Philippine asphalt national roads. Using correlation and multiple regression analysis on pavement photographs evaluated by DPWH district offices nationwide, the researchers established a locally validated PCI model with an R^2 value of 0.58. The study also noted that the current DPWH Visual Condition Index (VCI) lacks sufficient localized justification in existing literature. Their research provides a standardized local baseline for evaluating pavement conditions, which is essential for future automated road distress assessment systems.

A more recent implementation was conducted by Bustria et al. (2025), who developed a YOLOv11-based Android road damage detection application for Baguio City. The study demonstrated that deep learning technologies can effectively detect potholes and road defects using mobile devices. However, the researchers identified limitations such as battery consumption, thermal throttling, and inconsistent processing performance during prolonged monitoring operations. These findings support the growing preference for centralized web-based processing systems rather than relying solely on mobile-device computation.

The reviewed local studies collectively demonstrate that although the Philippines faces considerable infrastructure, environmental, and technological challenges, advancements in AI-based road damage detection, YOLO architectures, transfer learning, geospatial technologies,

GPS synchronization, and localized pavement assessment models provide promising solutions for improving automated road monitoring and maintenance workflows in the country.

2.3 Synthesis

The reviewed literature and studies demonstrate the rapid advancement of artificial intelligence, deep learning, and geospatial technologies in the field of automated road damage detection. Foreign studies emphasized the effectiveness of YOLO-based architectures, particularly YOLOv8 and other advanced object detection models, in detecting potholes and pavement defects with high speed and accuracy. Researchers highlighted the importance of feature extraction improvements, attention-enhanced learning, transfer learning, and large-scale datasets in improving detection reliability across different road environments and pavement conditions.

The literature also revealed the growing importance of centralized processing systems in infrastructure monitoring. Studies showed that server-side and web-based processing improve scalability, monitoring efficiency, and data organization while reducing the hardware limitations commonly encountered in mobile devices during continuous deep learning operations. Furthermore, integrating GPS synchronization and geospatial mapping technologies enables more accurate localization of road damage, improves route-based monitoring, and supports more efficient infrastructure maintenance planning. Local studies provided insight into the challenges of implementing automated road monitoring systems within the Philippine context. Factors such as tropical weather conditions, flooding, irregular road textures, pothole variability, environmental clutter, and inconsistent lighting significantly affect detection accuracy when using models trained only on foreign datasets. Because of these conditions, local researchers

emphasized the importance of localized datasets, transfer learning, GPS integration, localized pavement assessment models, and centralized monitoring systems for improving road inspection efficiency in Philippine environments. The reviewed studies also demonstrated that deep learning and YOLO-based systems can effectively detect road damage using locally trained datasets and transfer learning approaches. However, several limitations remain, including inconsistent detection performance caused by varying pavement conditions, hardware limitations in mobile-device implementations, battery consumption, thermal throttling, and the continued dependence on manual road inspection practices in the Philippines. These findings further support the need for scalable and localized automated monitoring systems specifically designed for Philippine infrastructure conditions. Although existing studies successfully implemented AI-based road monitoring technologies, there remains limited research focused on combining YOLOv8-based video analysis, GPS synchronization, geospatial visualization, and centralized web-based processing into a unified road damage detection platform specifically designed for Philippine LGUs and infrastructure agencies. Therefore, the present study aims to address this research gap by developing a web-based road damage detection and mapping system capable of analyzing uploaded road survey videos, synchronizing detections with GPS track data, and generating interactive geospatial visualizations to support organized, scalable, and localized infrastructure monitoring workflows.

2.3.1 Synthesis Table

Author/Year	Focus of study	Methodology	Key Findings	Synthesis/ Implication
Foreign studies				
Redmon et al. (2016)	Development of the YOLO real-time object detection framework	Introduced a unified convolutional neural network for real-time object detection	YOLO achieved fast and accurate object detection suitable for real-time applications	Provides the foundational object detection architecture used in the proposed YOLOv8-based road damage detection system
Arya et al. (2022)	Development of the RDD2022 road damage dataset	Compiled and annotated multinational road damage images for model training	Improved dataset diversity and strengthened model generalization across varying road conditions	Serves as the primary dataset source for training and evaluating the proposed system
Pham et al. (2022)	Real-time road damage detection using deep learning	Applied YOLO-based image analysis for detecting potholes and road cracks	YOLO models effectively identified road defects from images and videos	Supports the use of deep learning for automated road damage analysis

Roy & Bhaduri (2023)	Real-time road damage detection using YOLOv8	Implemented YOLOv8 for detecting potholes and road anomalies	YOLOv8 improved detection speed and processing performance	Demonstrates the effectiveness of YOLOv8 for road damage detection
Pham et al. (2024)	Improvements in YOLO architectures for road anomaly detection	Evaluated successive YOLO models for road defect detection	Improved feature extraction and multi-scale detection accuracy	Supports the selection of YOLOv8 as a reliable detection model
Irsal et al. (2025)	Real-time pothole detection and localization	Implemented YOLOv8 for detecting and localizing potholes from road imagery	YOLOv8 achieved accurate localization and efficient real-time processing	Validates the use of YOLOv8 in the proposed detection workflow
Yin et al. (2024)	Detection of road hazards using YOLOv8	Applied multi-scale feature learning for pothole and road hazard detection	Achieved high mean Average Precision (mAP) results	Supports the capability of YOLOv8 to detect multiple road damage types
Zeng & Zhong (2024)	Improvement of YOLOv8 for pavement	Developed the YOLOv8-PD model with	Reduced false detections and improved	Highlights the importance of customized

	damage detection	optimized feature fusion	detection reliability	YOLOv8 architectures
Zhang et al. (2025)	Small-object pothole detection using OBC-YOLOv8	Integrated attention-enhanced feature learning into YOLOv8	Improved detection of distant and low-resolution potholes	Relevant for analyzing dashcam-based road survey videos
iWatchRoad Project (2026)	Web-based pothole detection and geospatial visualization	Developed a scalable web platform using dashcam uploads and mapping tools	Centralized systems improved accessibility and road monitoring efficiency	Closely related to the proposed web-based detection and mapping workflow
Local studies				
Sarmiento (2021)	Pavement distress detection on Philippine roads	Applied YOLOv4 and DeepLabv3 on locally collected road images	Foreign-trained models struggled with Philippine road conditions	Emphasizes the need for localized datasets and model adaptation
Sarmiento & Reyes (2022)	Deep learning-based road damage classification for Philippine highways	Utilized convolutional neural networks for pothole and crack classification	Locally trained datasets improved detection performance	Supports the importance of localized model

Fortin et al. (2024)	EyeWay pothole mapping system	Combined computer vision and GPS technologies for pothole mapping	Geospatial technologies improved pothole localization	Supports GPS synchronization and mapping integration
Bustria et al. (2025)	YOLOv11-based Android road damage detection application	Developed a mobile road damage detection system for Baguio City	Mobile processing introduced hardware and performance limitations	Supports the need for centralized web-based processing instead of mobile-only systems
Manalo et al. 2022	Transfer learning-based pothole detection system for Philippine roads	Applied YOLOv3 with transfer learning on a locally compiled pothole image dataset; developed at Batangas State University, Adamson University, and FEU Institute of Technology	Achieved 95.43% average precision; accuracy in identifying individual potholes ranged from 33–69%, reflecting variability in pothole shapes and sizes on local roads	Demonstrates the feasibility of using pre-trained deep learning models for pothole detection in the Philippine context with limited local training data
Dacanay et al. 2024	Assessment of road condition monitoring	Reviewed and compared current DPWH	The Philippines remains reliant	Provides a policy-level foundation for

	practices and technologies in the Philippines	road monitoring practices against international automated systems; conducted under DOST-funded Project PAVE at UP Diliman	on manual inspection; existing international automated systems are not directly applicable due to local road conditions and resource constraints	developing automated road survey equipment tailored to Philippine road infrastructure needs
Ramos et al. 2024	Development of a Pavement Condition Index for Philippine asphalt national roads	Used correlation and multiple regression analysis on expert-evaluated pavement photos from DPWH district offices nationwide to derive a localized PCI model	The PCI model achieved an R^2 of 0.58; the current DPWH Visual Condition Index (VCI) lacks localized justification in published literature	Establishes a locally validated pavement condition rating baseline critical for standardizing automated distress assessment in the Philippines

CHAPTER III

RESEARCH METHODOLOGY

3.1 System Development Methodology



Agile Development Methodology is a flexible and iterative software development approach that focuses on continuous improvement, collaboration, and incremental delivery of system features. Unlike traditional development models that complete the entire project in a single cycle, Agile divides the development process into smaller phases called sprints. Each sprint produces a working component of the system that can be tested, evaluated, and improved based on feedback. Agile emphasizes adaptability, user involvement, rapid development, and continuous monitoring, making it highly suitable for projects that require frequent refinement and integration of multiple technologies such as computer vision, GPS synchronization, video analysis, and geospatial mapping.

For this study, Agile methodology is appropriate because the proposed web-based road damage detection and mapping system involves several interconnected modules, including video uploading, YOLOv8-based detection, GPS synchronization, timestamp extraction, database

management, and interactive map visualization. Through Agile development, the researchers can gradually design, develop, test, and improve each module while continuously evaluating detection accuracy, synchronization performance, usability, and overall system functionality. The iterative nature of Agile also allows the researchers to address technical challenges such as video quality variations, GPS inaccuracies, and detection inconsistencies during development.

3.1.1 Phase 1: System Foundation and Core Development

The first phase served as the foundational stage where the primary architecture and core functionalities of the proposed system were established. This phase focused on translating the conceptual framework into an operational prototype capable of processing uploaded road survey videos and GPS files.

This phase involved:

- Designing the initial frontend interface for uploading road survey videos and GPS track files
- Configuring the backend processing environment
- Collecting and preprocessing the RDD2022 dataset and locally collected road images
- Preparing annotations for potholes, longitudinal cracks, transverse cracks, and alligator cracks
- Implementing the initial YOLOv8 object detection model
- Developing the first frame extraction and timestamp processing workflow
- Establishing preliminary GPS synchronization functionality

Initial testing revealed limitations in detection consistency, timestamp extraction, and geographic synchronization accuracy, indicating the need for further refinement and optimization.

3.1.2 Phase 2: Detection Refinement and Feature Expansion

Detection refinement and feature expansion involved improving the reliability and precision of the system's processing capabilities. This phase focused on refining machine learning performance and expanding the system's ability to handle varying environmental and video conditions more effectively.

Enhancements included:

- Retraining the YOLOv8 model using expanded datasets and local road images
- Improving image preprocessing and augmentation techniques
- Refining frame extraction intervals and processing logic
- Improving bounding box generation and confidence threshold handling
- Enhancing timestamp extraction accuracy
- Expanding GPS synchronization workflows
- Improving frontend responsiveness and upload handling

This iteration significantly improved detection relevance, synchronization consistency, and overall system performance.

3.1.3 Phase 3: Geospatial Integration and Data Synchronization

Geospatial integration and data synchronization focused on embedding mapping and location-based functionalities into the system to support organized road monitoring workflows. The goal of this phase was to connect automated detection outputs with synchronized geographic visualization.

This phase introduced:

- GPS synchronization between detection timestamps and GPX or CSV files
- Automated geographic coordinate extraction
- Interactive geospatial visualization using React-Leaflet and OpenStreetMap
- Road damage marker generation and route visualization
- Improved management of synchronized road damage records
- Enhanced filtering and categorization of detected road damages
- Better alignment between detection outputs and mapped geographic locations

These refinements improved the system's ability to organize and visualize road damage information more effectively.

3.1.4 Phase 4: System Integration and Interface Enhancement

System integration and interface enhancement focused on combining all developed modules into a centralized and fully operational web-based application. This phase emphasized improving usability, accessibility, and interaction between the user and the automated analysis system.

New features included:

- Integration of frontend and backend functionalities
- Improved upload workflows for large video files
- Enhanced interface organization and visual hierarchy
- Better navigation for map visualization and damage records
- Dynamic loading of geographic markers and detection outputs
- Improved asynchronous processing workflows
- Enhanced display of timestamps, coordinates, and damage classifications

This phase increased the operational efficiency and usability of the proposed application.

3.1.5 Phase 5: Optimization and Validation

This phase served as the stabilization stage where the system underwent continuous refinement to ensure technical reliability, processing efficiency, and overall usability. The system transitioned from a developmental prototype into a validated platform ready for testing and evaluation.

Activities included:

- Frontend polishing and interface refinement
- Minor bug fixes and synchronization corrections
- Detection result formatting improvements
- Database optimization
- GPS mapping validation
- Processing performance optimization

- Functionality testing
- Detection accuracy evaluation
- Final system validation

This phase prepared the proposed system for testing, evaluation, and deployment.

3.1.6 Future Development Considerations

While the proposed system currently operates as a web-based platform utilizing uploaded road survey videos and GPS track files, future iterations of the system may incorporate additional intelligent infrastructure monitoring functionalities.

Future enhancements may include:

- Real-time roadside detection through mobile devices
- Live GPS synchronization and cloud integration
- Integration with drone-based road inspection systems
- Expansion of detectable road damage categories
- Automated road damage severity assessment
- Maintenance prioritization and reporting tools
- Multi-user collaboration and monitoring features

The development of larger locally collected Philippine road datasets was also identified as a significant opportunity for improving detection compatibility and model accuracy under varying local road conditions.

3.2 Conceptual Framework

3.2.1 Conceptual Framework Diagram

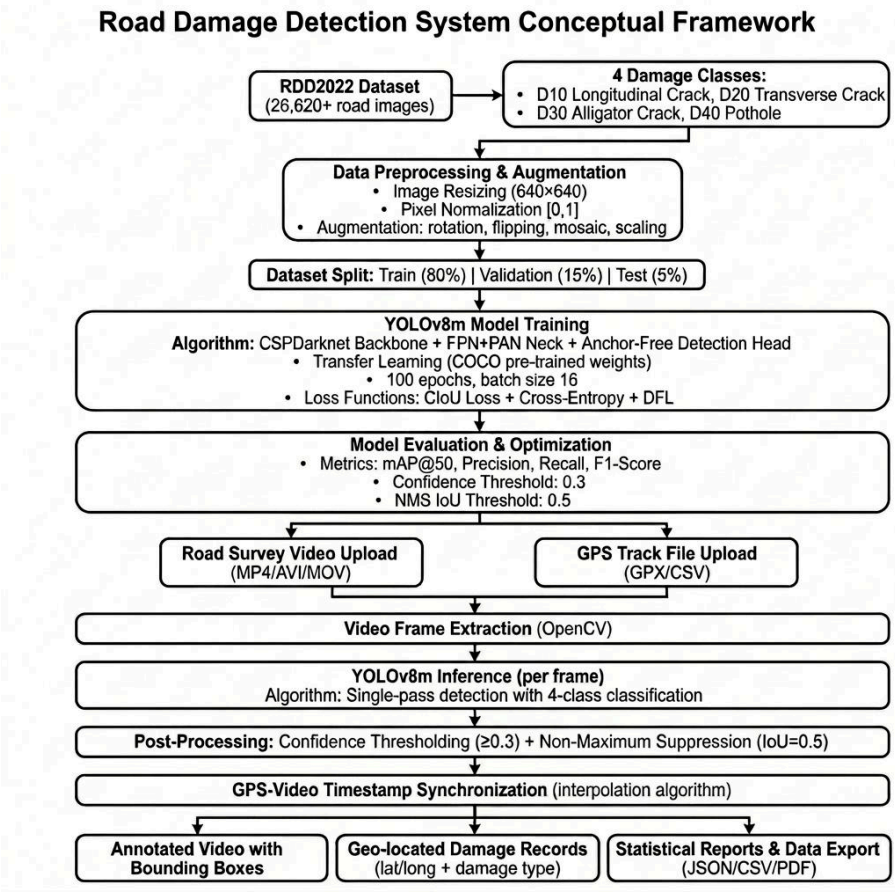


Figure 3.2.1 Road Damage Detection System Conceptual Framework

The conceptual framework for the Road Damage Detection System represents the complete development vision of the system as a whole—from data collection and model training to the deployment of a functional web-based application that serves road maintenance personnel, engineers, and infrastructure administrators. This framework establishes the theoretical and technical foundation for building an integrated solution that automates the identification, classification, and geo-location of road surface damages from survey video footage, ultimately improving the process of documenting, locating, and monitoring road damage.

The detailed flow diagram above illustrates the entire end-to-end process of the system's development and operation, specifying the algorithms and techniques used at each stage:

Data Collection — The system begins with the RDD2022 (Road Damage Dataset 2022), a comprehensive collection of over 26,620 annotated road images sourced from multiple countries with diverse road conditions. The dataset is organized into four damage classes based on the standardized road damage classification scheme: D10 (Longitudinal Crack), D20 (Transverse Crack), D30 (Alligator Crack), and D40 (Pothole). This multi-class dataset provides the labeled training examples required for supervised learning.

Data Preprocessing and Augmentation — Collected images undergo preprocessing to standardize inputs for the deep learning model. All images are resized to 640×640 pixels (the required input dimensions for YOLOv8m), and pixel values are normalized to the [0, 1] range using the formula $N(x,y) = I(x,y) / 255.0$. Data augmentation techniques—including rotation, horizontal flipping, mosaic augmentation, and scaling—are applied to artificially expand the training set and improve model robustness to variations in lighting, angle, and road surface texture. The dataset is then split into training (80%), validation (15%), and test (5%) subsets to enable proper model development and unbiased evaluation.

YOLOv8m Model Training — The core detection algorithm is YOLOv8m (You Only Look Once, version 8, medium variant), which implements a three-component architecture: the CSPDarknet backbone for multi-scale feature extraction, the FPN+PAN (Feature Pyramid Network + Path Aggregation Network) neck for feature fusion across scales, and an anchor-free detection head that directly predicts bounding box coordinates

and class probabilities. Training employs transfer learning from COCO pre-trained weights to leverage general object detection knowledge, running for 100 epochs with a batch size of 16. The model optimizes three loss functions simultaneously: CIoU (Complete Intersection over Union) Loss for bounding box regression accuracy, Cross-Entropy Loss for damage class classification, and DFL (Distribution Focal Loss) for fine-grained localization.

Model Evaluation and Optimization — The trained model is evaluated against the held-out test set using standard object detection metrics: mAP@50 (mean Average Precision at IoU threshold 0.5), Precision, Recall, and F1-Score. Based on the evaluation results, the confidence threshold is calibrated (default 0.3, representing the minimum detection certainty required) and the NMS IoU threshold is set to 0.5 (overlap threshold for duplicate removal), balancing detection sensitivity against false positive rates. The final model performance metrics are presented and discussed in the Results chapter.

System Input — In operation, the system accepts two concurrent inputs from the user: a road survey video file (MP4, AVI, or MOV format) captured during field inspection, and a GPS track file (GPX or CSV format) recorded simultaneously with the video. These paired inputs enable the system to link visual detections to geographic coordinates.

Video Processing Pipeline — The uploaded video is processed through a sequential pipeline using the trained YOLOv8m model. First, OpenCV's VideoCapture extracts individual frames from the video. Each frame is then passed through the YOLOv8m inference engine, which performs single-pass detection with 4-class classification—simultaneously predicting bounding box locations, damage type classifications, and confidence scores. Post-processing applies confidence thresholding (≥

0.3) to filter low-certainty predictions and Non-Maximum Suppression (IoU = 0.5) to eliminate duplicate detections for the same damage instance. Finally, GPS-Video Timestamp Synchronization maps each detection to real-world coordinates by interpolating the GPS track data against the video frame timestamp.

System Output — The pipeline produces three categories of output: annotated video with color-coded bounding boxes showing damage locations and classifications; geo-located damage records containing GPS coordinates (latitude/longitude) paired with damage type and confidence for each detection; and statistical reports with data export capabilities in JSON, CSV, and PDF formats for use by road maintenance personnel, engineers, and infrastructure administrators in planning repair operations and prioritizing maintenance.

3.2.2 System Architecture Block Diagram

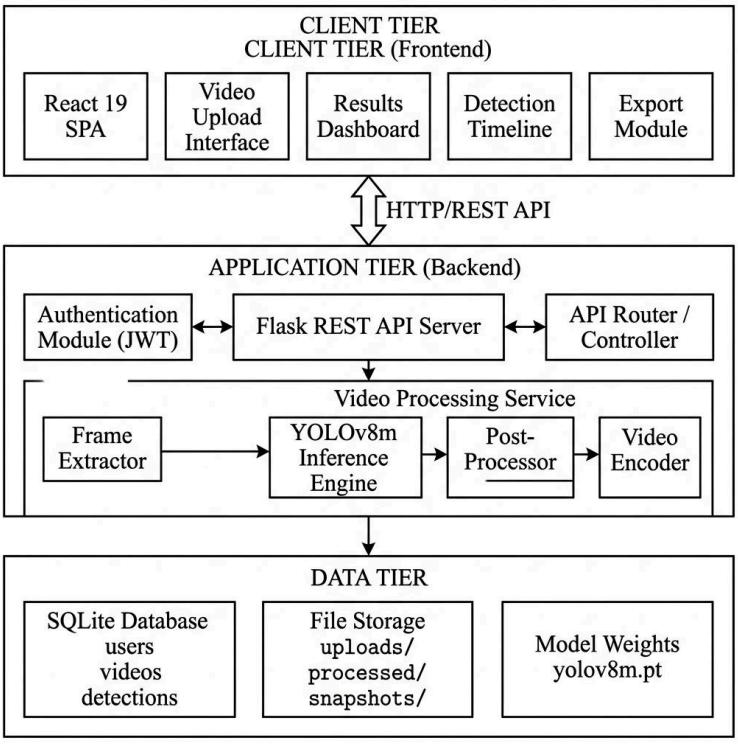


Figure 3.2.2 Three-Tier System Architecture

The system architecture follows a three-tier design pattern that separates concerns across distinct layers, enabling independent development, testing, and scaling of each component.

Client Tier (Frontend) — Built on React 19 as a single-page application (SPA), this tier provides the user interface through which road maintenance personnel, engineers, and infrastructure administrators interact with the system. Key modules include: the Video Upload Interface for submitting road survey footage and GPS track files; the Results Dashboard for viewing detection outcomes; the Detection Timeline for navigating through detected damages chronologically; and the Export Module for downloading reports. The frontend communicates with the backend exclusively through HTTP/REST API calls, ensuring loose coupling between presentation and business logic.

Application Tier (Backend) — The Flask REST API Server serves as the central processing hub. The Authentication Module implements JWT (JSON Web Token) bearer token security for API access control. The API Router/Controller dispatches incoming requests to appropriate handlers. The Video Processing Service is the computational core, containing four sequential sub-components: the Frame Extractor (using OpenCV to decompose video into individual frames), the YOLOv8m Inference Engine (executing the deep learning model on each frame), the Post-Processor (converting raw model predictions into structured detection records with bounding boxes, confidence scores, and GPS coordinates through timestamp synchronization), and the Video Encoder (re-encoding annotated frames into a playable output video). Video

processing runs asynchronously in background threads to prevent blocking user interactions.

Data Tier — This tier manages all persistent storage through three components: the SQLite Database stores relational data including user accounts, video metadata, detection records with GPS coordinates, and GPS track data; the File Storage system organizes uploaded videos (uploads/), GPS track files, processed annotated videos (processed/), and individual detection frame captures (snapshots/) in structured directories; and the Model Weights storage maintains the YOLOv8m pre-trained model file (yolov8m.pt) for inference operations.

3.3 Algorithm and Its Rules

3.3.1 YOLOv8m Deep Learning Algorithm

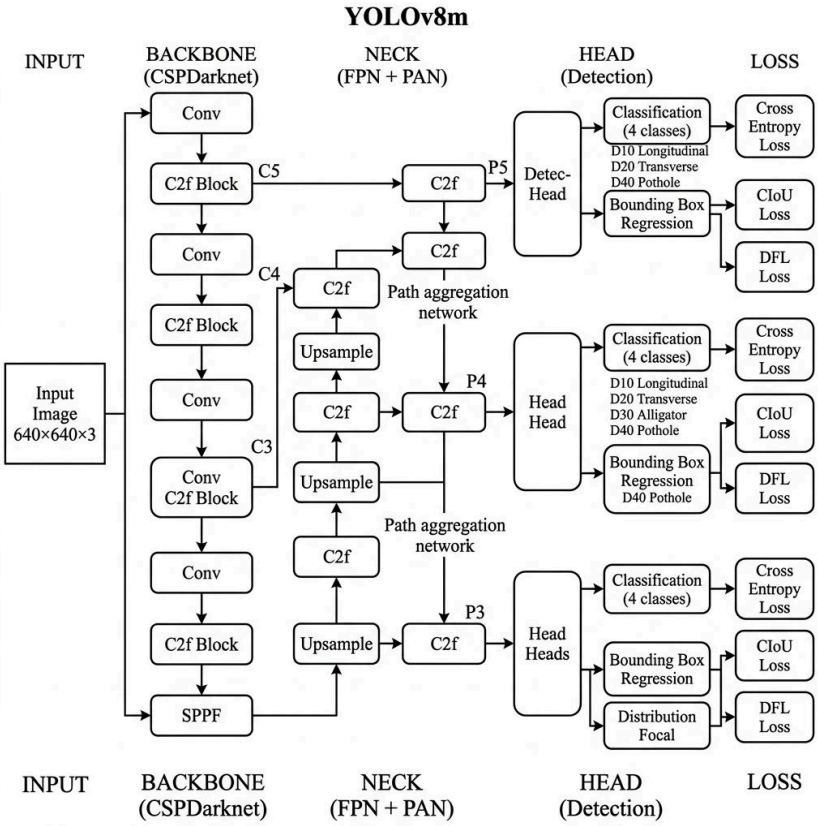


Figure 3.3.1 YOLOv8m Architecture Diagram

The You Only Look Once (YOLO) algorithm is one of the most widely adopted object detection frameworks in computer vision research and applications. The researchers implemented YOLOv8m (medium variant) for the Road Damage Detection System, which improves upon previous YOLO iterations with enhanced architecture and performance. This algorithm represents a fundamental advancement in computer vision that enables single-stage detection—predicting bounding boxes and class probabilities simultaneously in a single forward pass through the network (Jocher et al., 2023). This unified approach enables significantly faster processing compared to traditional two-stage detectors, making it suitable for processing large volumes of road survey video footage. The system benefits road maintenance personnel, engineers, and infrastructure administrators by automating what has traditionally been a manual, time-consuming, and subjective inspection process.

The researchers preprocess each input frame to standardized dimensions of 640×640 pixels. The normalized image is then passed through feature extraction algorithms in the backbone network (CSPDarknet), which employs cross-stage partial connections to enhance information flow while reducing computational requirements. Feature pyramid algorithms in the neck network then perform multi-scale feature fusion using path aggregation techniques. The detection head implements anchor-free prediction algorithms rather than using predefined anchor boxes. This algorithmic approach makes the model more adaptable to varying shapes and sizes of road damage. The algorithm employs non-maximum suppression to eliminate redundant detections, using a greedy algorithm that iteratively selects boxes with maximum confidence and removes boxes with significant overlap.

Steps in Algorithm:

1. Preprocess video frame to 640×640 pixels for consistency.
2. Normalize pixel values to [0, 1] range.
3. Run the image through the YOLOv8m model for damage identification.
4. Extract bounding boxes with confidence scores for four damage classes.
5. Apply Non-Maximum Suppression to remove overlapping detections.
6. Synchronize detections with GPS coordinates via timestamp matching.
7. Return filtered detections above confidence threshold (≥ 0.3).

3.3.2 Formulas and Mathematical Representations

3.3.2.1 Time Complexity and Pixel Normalization

Time Complexity

1. Model Training:

$$O(ebi) \tag{1}$$

where:

- e = number of epochs (100)
- b = number of batches (16)
- i = number of images per batch

2. Model Inference:

$$O(p) \tag{2}$$

where:

- p = number of pixels in the input image (640×640 = 409,600 pixels)

3. Non-Maximum Suppression:

$$O(n^2) \tag{3}$$

where:

- n = number of initial detections before filtering

Mathematical Representation

1. Image Preprocessing (Pixel Normalization):

$$N(x,y) = \frac{I(x,y)}{255.0}. \tag{4}$$

where:

- $N(x,y)$ = normalized pixel value at position (x,y)
- $I(x,y)$ = original pixel intensity at position (x,y)

Figure 3.3.2a: Time Complexity and Normalization Formulas

Time Complexity:

Formula (1): Model Training Complexity — O(ebi)

The training time complexity is determined by the product of three factors: the number of epochs e (training iterations over the full dataset, set to 100), the number of batches b per epoch (set to 16), and the number of images i processed per batch. This represents the total number of forward and backward passes through the neural network during the training phase. Each pass involves computing gradients and updating approximately 25.9 million model parameters.

Formula (2): Model Inference Complexity — O(p)

During inference, the computational cost scales linearly with the number of pixels p in the input image. With the standardized input size of 640×640 pixels (409,600 total pixels across 3 RGB channels), this linear relationship ensures predictable processing times. The convolutional operations slide fixed-size kernels across the image, processing each spatial location once per layer.

Formula (3): Non-Maximum Suppression Complexity — $O(n^2)$

NMS has quadratic time complexity because each detection must be compared against every other detection to calculate IoU (Intersection over Union) overlap. In practice, n is small (typically fewer than 100 detections per frame after confidence filtering), so this step adds minimal overhead to the overall pipeline.

Mathematical Representation:

Formula (4): Image Preprocessing (Pixel Normalization) — N(x,y) = I(x,y) / 255.0

The normalization step scales all pixel intensity values from the integer range [0, 255] to the floating-point range [0.0, 1.0]. This standardization ensures consistent input distribution for the neural network, which improves convergence during training and stability during inference. Each pixel across all three RGB channels is independently divided by 255.0, the maximum possible intensity value.

3.3.2.2 IoU, Confidence, and Evaluation Metrics

2. Intersection over Union (IoU):

$$\text{IoU} = \frac{A_{\text{overlap}}}{A_{\text{union}}} = \frac{A_1 \cap A_2}{A_1 \cup A_2} \tag{5}$$

where:
- A_1 = area of predicted bounding box
- A_2 = area of ground truth bounding box
- A_{overlap} = intersection area between boxes
- A_{union} = union area of both boxes

3. Confidence Scores:

$$\text{Conf}_i = P_{\text{obj},i} \times \text{IoU}_i \tag{6}$$

4. Evaluation Metrics:

Precision:

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{7}$$

Recall:

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{8}$$

Mean Average Precision:

$$\text{mAP} = \int_0^1 p(r)\, dr \tag{9}$$

F1-Score:

$$\text{F1} = 2 \times \frac{P \times R}{P + R} \tag{10}$$

where:
- P = precision
- R = recall
- mAP = mean Average Precision
- TP = true positives
- FP = false positives
- FN = false negatives

Figure 3.3.2b: IoU, Confidence, and Evaluation Metric Formulas

Formula (5): Intersection over Union (IoU)

IoU is the primary metric used to determine whether a predicted bounding box sufficiently overlaps with the ground truth annotation. It is computed as the ratio of the intersection area (where both boxes overlap) to the union area (the total area covered by either box). A higher IoU indicates better localization accuracy. In the system's NMS step, any two detections of the same class with IoU > 0.5 are considered duplicates, and the lower-confidence detection is discarded. During evaluation, a detection is considered a true positive only if its IoU with the closest ground truth box exceeds 0.5 (the mAP@50 standard).

Formula (6): Confidence Score — $Conf_i = P_{obj,i} \times IoU_i$

The confidence score is a composite metric that combines two factors: the objectness score P_{obj} (how likely the region contains any object) and the predicted IoU (how well the bounding box fits the detected object). This dual-factor approach ensures that only detections that are both confidently identified and well-localized receive high scores, reducing false positives in the output.

Formula (7): Precision — $P = TP / (TP + FP)$

Precision measures the reliability of the system's predictions. A precision of 60.5% means that approximately 6 out of every 10 detections flagged by the system correspond to actual road damage. High precision is important for road maintenance personnel who rely on the system's outputs to plan repair operations—false positives waste inspection resources.

Formula (8): Recall — R = TP / (TP + FN)

Recall measures the system's sensitivity. A recall of 50.6% means the system successfully identifies roughly half of all actual road damages present in the video. For infrastructure administrators responsible for comprehensive road condition assessment, recall indicates the thoroughness of the automated survey.

Formula (9): Mean Average Precision — mAP = $\int_0^1$ p(r) dr

The mAP@50 score of 53.9% represents the overall detection quality across all four damage classes at the IoU threshold of 0.5. It is computed by integrating the precision-recall curve—the area under this curve captures the trade-off between precision and recall across all possible confidence thresholds, providing a single comprehensive quality metric.

Formula (10): F1-Score — F1 = 2 × (P × R) / (P + R)

The F1-Score provides a single balanced metric that harmonizes precision and recall through their harmonic mean. An F1-Score of approximately 0.556 indicates a balanced performance where neither precision nor recall dominates, which is desirable for a system that must both avoid false alarms and detect as many damages as possible.

3.3.2.3 Bounding Box Transformation, Letterboxing, GPS Synchronization, and Frame Rate

5. **Bounding Box Coordinate Transformation:**

$$x_{\min} = x_c - w/2 \tag{11}$$
$$y_{\min} = y_c - h/2 \tag{12}$$
$$x_{\max} = x_c + w/2 \tag{13}$$
$$y_{\max} = y_c + h/2 \tag{14}$$

where:
- (x_c, y_c) = center coordinates of the bounding box
- (w, h) = width and height of the bounding box

6. **Letterbox Scaling (Aspect Ratio Preservation):**

$$scale = \min(T_w/I_w, T_h/I_h) \tag{15}$$
$$pad_x = (T_w - I_w \times scale)/2 \tag{16}$$
$$pad_y = (T_h - I_h \times scale)/2 \tag{17}$$

where:
- T_w, T_h = target dimensions (640×640)
- I_w, I_h = original image width and height

7. **GPS-Video Timestamp Synchronization:**

$$t_{video}(f) = \frac{f}{\text{FPS}} \tag{18}$$
$$\text{GPS}(f) = \text{interpolate}(G, t_{video}(f) + t_{offset}) \tag{19}$$

where:
- f = frame number
- FPS = video frame rate
- G = GPS track point sequence
- t_{offset} = time offset between video and GPS recording

8. **Frame Processing Rate:**

$$FPS_{effective} = \frac{1000}{T_{frame}} \tag{20}$$
$$T_{total} = \frac{N_{frames} \times T_{frame}}{1000} \tag{21}$$

Figure 3.3.2c: Bounding Box, Letterboxing, GPS Synchronization, and Frame Rate Formulas

Formulas (11)-(14): Bounding Box Coordinate Transformation

YOLOv8 internally represents bounding boxes using center-point format (x_center, y_center, width, height), which is natural for the regression task in the detection head. However, for rendering bounding boxes on video frames and storing them in the database, the system converts to corner-point format (x_min, y_min, x_max, y_max). This transformation is essential for the OpenCV drawing

functions used in the annotation pipeline and for providing human-readable coordinate data in exported reports.

Formulas (15)-(17): Letterbox Scaling (Aspect Ratio Preservation)

Letterboxing ensures that the input image is resized to the model's expected 640×640 input without distortion. The image is scaled uniformly by the smaller of the two axis ratios (Formula 15), and the remaining space is filled with padding (Formulas 16-17). After inference, the padding offset must be subtracted from the predicted coordinates, and the scale factor must be reversed to map detections back to the original image dimensions. This is critical for ensuring that bounding box coordinates accurately correspond to actual damage locations in the original video frames.

Formulas (18)-(19): GPS-Video Timestamp Synchronization

The GPS synchronization formulas enable the system to assign geographic coordinates to each detected road damage. Formula (18) converts a frame number f to a video timestamp by dividing by the frame rate (FPS). Formula (19) then performs linear interpolation on the GPS track point sequence G to find the geographic position corresponding to that video timestamp, accounting for a time offset t_offset between when the video recording and GPS logging were started. This interpolation is necessary because GPS track points are typically recorded at 1 Hz (one per second) while video frames are captured at 24-30 Hz, so exact timestamp matches are rare. The resulting latitude and longitude coordinates allow each detection to be plotted on a map, enabling engineers and infrastructure

administrators to precisely locate road damages for maintenance dispatch.

Formulas (20)-(21): Frame Processing Rate

These formulas quantify the system's throughput. With an average inference time of approximately 200ms per frame on CPU, the system processes roughly 5 frames per second (Formula 20). For a typical 60-second road survey video captured at 30 fps, the total frame count is 1,800, yielding an estimated processing time of approximately 6 minutes (Formula 21). These metrics are critical for setting user expectations through the progress tracking interface.

3.4 Operational Framework

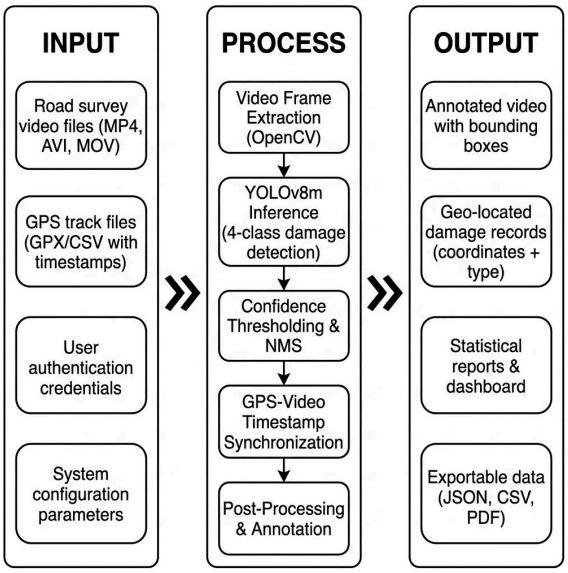


Figure 3.4 IPO (Input-Process-Output) Operational Framework

The researchers implemented a comprehensive operational framework organized as an IPO (Input-Process-Output) model, which structures system operations into three distinct phases that ensure a clear, traceable, and repeatable workflow from raw video input to geo-located, classified road damage output. This framework benefits road maintenance personnel, engineers,

and infrastructure administrators by transforming unstructured road survey footage into structured, actionable assessment data.

Phase 1: Input

The input phase establishes the foundation for the detection process through four primary input sources. The principal input is the road survey video file, supporting multiple formats including MP4, AVI, MOV, MKV, and WebM, with a maximum file size of 500 MB. These videos are typically captured by field personnel using phone or vehicle-mounted cameras during routine road surveys, capturing road surface conditions under diverse environmental contexts—varying lighting conditions, weather states, and road surface types. The second input source is the GPS track file, uploaded alongside the video in GPX or CSV format. This file contains timestamped geographic coordinates (latitude, longitude, elevation) recorded by a GPS logger or smartphone app running simultaneously with the video capture. The GPS track file is the key enabler for geo-located damage mapping—without it, detections can only be identified by video timestamp, but with it, every detection is assigned precise real-world coordinates. The third input source is the user authentication credentials, processed through the JWT (JSON Web Token) authentication module, which ensures that only authorized personnel can access the system and submit videos for analysis. The fourth input encompasses system configuration parameters, including the confidence threshold setting (default 0.3), the NMS IoU threshold (default 0.5), and output format preferences.

Phase 2: Process

The processing phase represents the computational core of the system, where raw video and GPS inputs are transformed into structured, geo-located detection data through a multi-stage pipeline. The first stage is Video Frame Extraction using OpenCV, which decomposes the

uploaded video into individual frames based on the video's native frame rate. Each extracted frame is normalized to the 640×640 pixel input size required by the YOLOv8m model using letterboxing to preserve the original aspect ratio. The second stage is YOLOv8m Inference (4-class damage detection), where each preprocessed frame passes through the deep learning model's backbone (CSPDarknet), neck (FPN+PAN), and detection head to produce raw predictions containing bounding box coordinates, class probabilities for the four damage types (D10 Longitudinal Crack, D20 Transverse Crack, D30 Alligator Crack, D40 Pothole), and objectness scores. The third stage applies Confidence Thresholding and NMS, filtering out all detections below the 0.3 confidence threshold and performing Non-Maximum Suppression to remove duplicate detections for the same damage instance. The fourth stage performs GPS-Video Timestamp Synchronization, which matches each detection's video frame timestamp to the corresponding GPS track point through interpolation, assigning latitude and longitude coordinates to every validated detection. This step transforms frame-level detections into geo-located damage records. The fifth and final processing stage is Post-Processing and Annotation, where validated detections are used to draw color-coded bounding boxes on the original frames, annotate damage type labels with confidence scores, capture individual detection snapshots, and re-encode the annotated frames into a playable output video file.

Phase 3: Output

The output phase delivers processed results through multiple channels. The primary output is the annotated video with bounding boxes, where each detected road damage is visually highlighted with its damage type label and confidence score, enabling personnel to visually verify detection results. The second output is geo-located damage records containing the damage type classification, confidence score, bounding box coordinates, GPS coordinates (latitude and

longitude), frame number, and processing timestamp for each detection. The third output is the statistical reports and interactive dashboard, which aggregates detection data into summary statistics including total detections by damage type, confidence score distributions, geographic distribution maps, and severity assessments. The fourth output encompasses exportable data in JSON (for programmatic integration), CSV (for spreadsheet analysis), and PDF (for formal reporting) formats, ensuring the data can be utilized across different organizational workflows by road maintenance personnel, engineers, and infrastructure administrators.

3.5 Use Case

The Road Damage Detection System is designed to streamline and automate the process of road damage identification, classification, and geo-location. The system serves a broad range of stakeholders involved in road infrastructure management: road maintenance personnel who conduct field surveys and coordinate repairs, engineers who assess structural damage severity and plan rehabilitation, and infrastructure administrators who oversee budgets, prioritize maintenance schedules, and manage reporting for organizational accountability. By leveraging YOLOv8m deep learning technology and GPS synchronization, the system improves the process of documenting, locating, and monitoring road damage from road survey videos.

The system serves two primary user roles: Road Maintenance Personnel (the primary users who capture, upload, and analyze road survey data) and System Administrators (who manage the platform, user accounts, and data integrity). Both groups interact with the application through a series of intuitive steps. The process begins at the login page, from which the authenticated user proceeds to upload a road survey video alongside its corresponding GPS track file. The system processes the video through the YOLOv8m detection pipeline and synchronizes detections with GPS coordinates. Once processing is complete, the system presents the annotated

video with detection results, a geo-located damage map, statistical summaries, and individual damage snapshots. Users can then analyze the detections by filtering by damage type (pothole, transverse crack, longitudinal crack, or alligator crack), sort by confidence level, view damages on a geographic map, and export the data in various formats for official reporting and infrastructure planning purposes.

3.5.1. Use Case Diagram

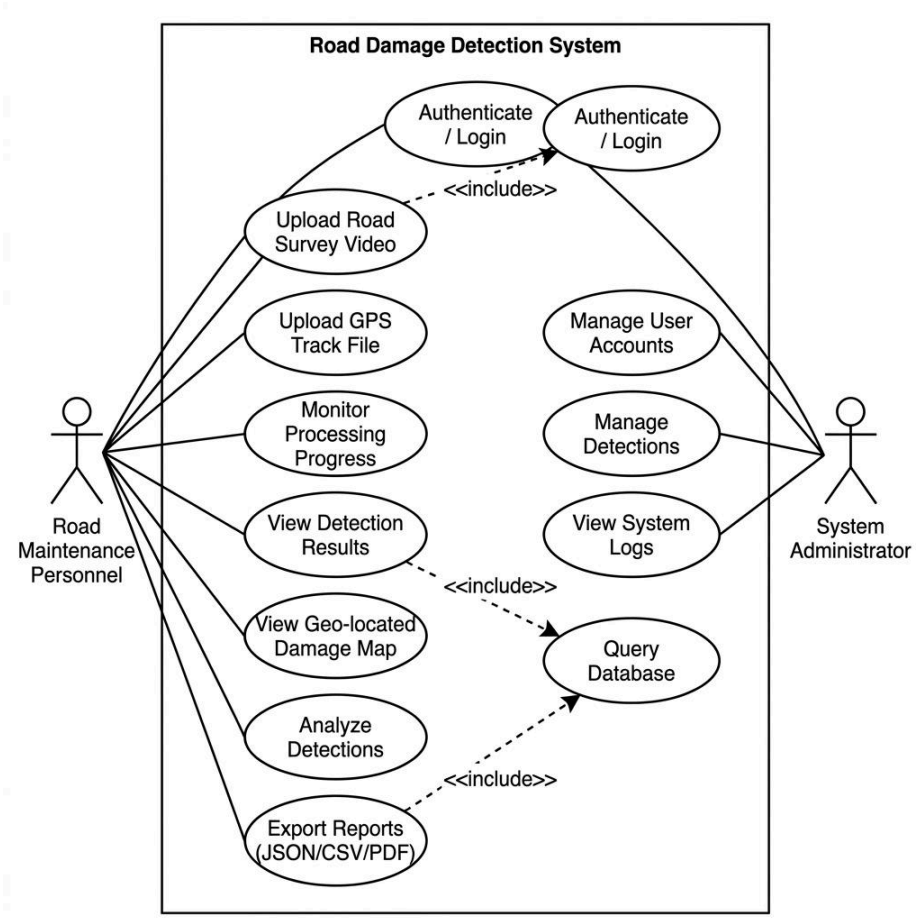


Figure 3.5.1 Use Case Diagram

The use case diagram illustrates the complete scope of system functionality and actor interactions using standard UML notation.

On the left, the Road Maintenance Personnel actor connects to eight use cases representing the primary user workflow: Authenticate/Login, Upload Road Survey Video,

Upload GPS Track File, Monitor Processing Progress, View Detection Results, View Geo-located Damage Map, Analyze Detections, and Export Reports (JSON/CSV/PDF). This actor represents the field personnel, engineers, and infrastructure administrators who use the system to document and assess road damage.

On the right, the System Administrator actor connects to four use cases for platform management: Authenticate/Login, Manage User Accounts, Manage Detections, and View System Logs.

The dashed arrows with <<include>> labels indicate mandatory dependencies: Upload Road Survey Video includes Authentication/Login (users must be authenticated before uploading); View Detection Results and Export Reports both include Query Database (data retrieval is an integral prerequisite for these operations). The system boundary rectangle encapsulates all use cases, clearly delineating the scope of the Road Damage Detection System.

3.6 Respondents of the Study

The respondents of this study will consist of selected personnel from Local Government Units (LGUs) and individuals involved in road monitoring, infrastructure maintenance, and transportation management. Specifically, the respondents may include municipal engineers, road maintenance personnel, infrastructure planners, administrative staff, and other authorized personnel responsible for road inspection and maintenance operations within the selected municipality or city.

The respondents will be chosen because they are the intended end-users and beneficiaries of the proposed web-based road damage detection and mapping system. Their roles involve monitoring road conditions, documenting road damage, coordinating maintenance activities,

and managing infrastructure-related records. As such, their evaluation and feedback are essential in determining the usability, efficiency, functionality, and practical applicability of the system in actual road maintenance workflows.

The study will utilize purposive sampling in selecting respondents, as only individuals directly involved in infrastructure management and road monitoring processes will participate in the evaluation of the system. The respondents will assess the system based on criteria such as usability, functionality, detection efficiency, mapping accuracy, accessibility of information, and overall usefulness in supporting road inspection and maintenance operations.

Additionally, information gathered from the respondents through surveys, interviews, or system evaluation forms will be used to analyze the effectiveness of the proposed system and identify possible improvements for future development.

3.7 Ethical Considerations

The development and deployment of the Road Damage Detection System is guided by ethical principles to ensure responsible use of technology, protection of data, and transparency in automated decision-making. As the system processes road survey video footage and GPS data collected from public road infrastructure, the researchers have identified and addressed the following ethical considerations.

Privacy, Anonymity, and Confidentiality

The system will be designed with the protection of privacy as a primary concern. Road survey videos captured by vehicle-mounted cameras may inadvertently record bystanders, vehicles with license plates, or private property adjacent to roadways. To mitigate privacy risks, the system processes video footage solely for the purpose of detecting road surface damage—specifically potholes, transverse cracks, longitudinal cracks, and alligator cracks. The

YOLOv8m model is trained exclusively on road damage classes and does not detect, identify, or classify people, vehicles, or personally identifiable features. No facial recognition, license plate recognition, or any form of biometric identification is performed at any stage of the processing pipeline. Video frames are analyzed only for road surface conditions, and the detection output contains only bounding box coordinates, damage classifications, confidence scores, and GPS coordinates—none of which constitute personally identifiable information. The GPS data collected pertains strictly to public road locations and does not track or identify individual users or residents. All uploaded video files and GPS track data are stored securely on the system's server with access restricted to authenticated users through JWT (JSON Web Token) bearer token authentication. Data is organized in user-specific contexts, ensuring that one user's uploaded survey data is not accessible to other users. The SQLite database stores detection records with anonymized reference identifiers, and no personal information beyond the user's registered account credentials is linked to detection data.

Informed Consent

All participants involved in the evaluation and testing of the Road Damage Detection System will be required to provide written informed consent before taking part in the study. Prior to signing the consent form, each participant will be given a thorough explanation of the study's purpose, the nature of the tasks they would perform (uploading videos, reviewing detections, completing evaluation questionnaires), the data that would be collected, and how that data would be used, stored, and eventually disposed of. Participants will be informed that road survey videos are processed by an automated deep learning model (YOLOv8m) for the sole purpose of road damage classification, and that no personal or biometric data is extracted or retained from the video footage. The consent form will explicitly state each participant's rights with respect to

their involvement in the study, including the right to access their data, request corrections, and request deletion.

Voluntary Participation

Participation in the study is voluntary, with no coercion or obligation to provide reasons for declining. Participants may withdraw at any time without penalty, and their data will be deleted upon request. Researchers will respect all decisions, ensuring participants have ample time to review consent forms and ask questions before starting. No incentives will be offered that could unduly influence participation decisions, and participants will not face any professional or personal consequences for choosing not to participate or for withdrawing mid-study.

No Deception

The study will involve no deception. Participants will be briefed on the system's function and the road damage detection process beforehand to ensure authentic engagement. This transparency is ethically and methodologically vital, as the evaluation measurements depend on genuine user responses and interactions with the system. Participants will be fully informed about: - The types of road damage the system detects (pothole, transverse crack, longitudinal crack, alligator crack) - How the YOLOv8m model processes video frames - How GPS synchronization maps detections to geographic coordinates - The purpose of any questionnaires or evaluation instruments administered - How their feedback will be used to improve the system

Data Handling and Retention

All road survey video files, GPS track data, and detection records collected during the study will be stored on secured, password-protected systems accessible only to the research team. Data will be retained only for the duration necessary to complete the research analysis and thesis documentation. Upon completion of the study, all video files and associated data will be

securely deleted or anonymized in accordance with institutional data management policies. No data will be shared with third parties without explicit written consent from participants.

Responsible Use of AI

The researchers acknowledge the inherent limitations of deep learning-based object detection models, recognizing that the system may produce both false positives (detecting damage where none exists) and false negatives (missing actual damage). The system is designed as a decision-support tool to assist—not replace—road maintenance personnel, engineers, and infrastructure administrators in their professional judgment. Detection results should be verified by qualified personnel before being used as the sole basis for maintenance decisions or budget allocations. The researchers commit to transparently reporting model performance metrics and limitations in all documentation and presentations.

3.8 Statistical Treatment

This section describes the statistical methods and tools used to analyze and interpret the data collected during the evaluation of the Road Damage Detection System. The statistical treatment is applied to both the model performance evaluation data and the user acceptance testing data.

3.8.1 Model Performance Evaluation

The YOLOv8m model's detection performance is evaluated using standard object detection metrics computed over the test dataset:

Mean Average Precision (mAP@50) — The primary metric for evaluating overall detection quality. mAP is computed by calculating the Average Precision (AP) for each of the four damage classes (D10, D20, D30, D40) independently, then averaging them. AP is defined as the area under the precision-recall curve at an IoU threshold of 0.5.

Precision — Measures the proportion of correct detections among all detections made by the model. Computed as TP / (TP + FP), where TP = true positives and FP = false positives.

Recall — Measures the proportion of actual road damages that the model successfully detected. Computed as TP / (TP + FN), where FN = false negatives.

F1-Score — The harmonic mean of precision and recall, providing a single balanced metric. Computed as 2 × (P × R) / (P + R).

Confusion Matrix — A matrix showing the distribution of predictions across all four damage classes, enabling identification of misclassification patterns (e.g., longitudinal cracks being classified as transverse cracks).

3.8.2 User Acceptance Testing

The system's usability and effectiveness are evaluated through user acceptance testing with road maintenance personnel, engineers, and infrastructure administrators. The following statistical methods are applied to the evaluation data:

Likert Scale Analysis — Evaluation questionnaires use a 5-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree) to measure user satisfaction across dimensions such as system usability, detection accuracy perception, interface design, and overall usefulness. Responses are analyzed using:

Weighted Mean ($\bar{x}_u$) — The average response score for each evaluation criterion, calculated as: $\bar{x}_u = \Sigma(f_i \times w_i) / N$ where f_i = frequency of responses at scale point i, w_i = weight of scale point i (1 through 5), and N = total number of respondents.

Standard Deviation (σ) — Measures the dispersion of responses around the weighted mean, indicating the degree of consensus among respondents: $\sigma = \sqrt{[\Sigma(x_i - \bar{x})^2 / (N - 1)]}$

Interpretation Scale for Weighted Mean:

Range	Verbal Interpretation
4.21 – 5.00	Strongly Agree / Excellent
3.41 – 4.20	Agree / Very Good
2.61 – 3.40	Neutral / Good
1.81 – 2.60	Disagree / Fair
1.00 – 1.80	Strongly Disagree / Poor

3.8.3 Frequency and Percentage Distribution

Descriptive statistics are used to summarize participant demographics and detection distribution data:

Frequency Distribution — The count of occurrences for each category (e.g., number of detections per damage type, number of respondents per role).

Percentage Distribution — The proportion of each category relative to the total, expressed as a percentage:

$$\text{Percentage} = (f / N) \times 100\%$$

where f = frequency of a particular category and N = total number of observations.

These distributions are used to present the breakdown of detected road damages by type (D10, D20, D30, D40), the distribution of confidence scores, and the demographic profile of evaluation participants.

3.8.4 Software Tools

The following software tools are used for statistical computation and analysis:

Tool	Purpose
Python (NumPy, Pandas)	Model performance metric computation, data aggregation
Ultralytics YOLOv8	Automated mAP, precision, recall, and F1 calculation during training/validation
Microsoft Excel / Google Sheets	Likert scale analysis, weighted mean calculation, chart generation
Matplotlib / Seaborn	Visualization of detection distributions, confusion matrices, precision-recall curves

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